



Long Division: 3-Digit $\div$ 1-Digit with Remainder





What is Long Division with Remainders?



Long division helps us divide large numbers by smaller ones. Sometimes the numbers don't divide evenly, and we get leftovers called remainders. This happens when the divisor doesn't fit perfectly into the dividend!





Understanding Remainders



A remainder is what's left over when numbers don't divide evenly. For example: Share 10 candies among 3 friends — each gets 3, and 1 candy is left!



Step-by-Step:

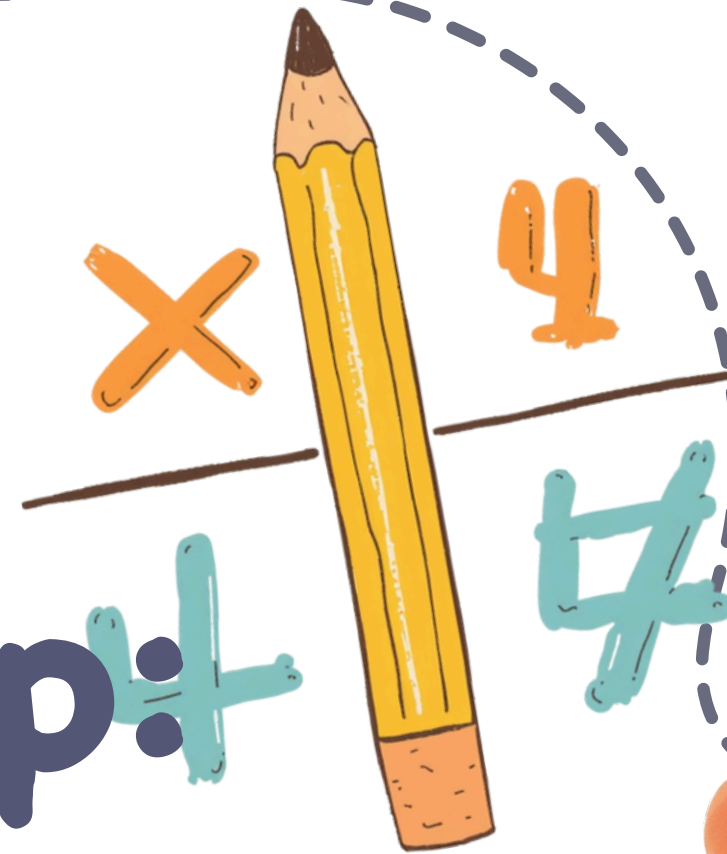
$456 \div 7$

1. Divide: $4 \div 7 = 0$, so try $45 \div 7 = 6$
2. Multiply: $6 \times 7 = 42$, subtract: $45 - 42 = 3$
3. Bring down 6 to get 36
4. Divide: $36 \div 7 = 5$, multiply: $5 \times 7 = 35$
5. Subtract: $36 - 35 = 1 \rightarrow$ Remainder is 1!
6. Answer: $456 \div 7 = 65 \text{ R}1$



Step-by-Step:

$789 \div 4$



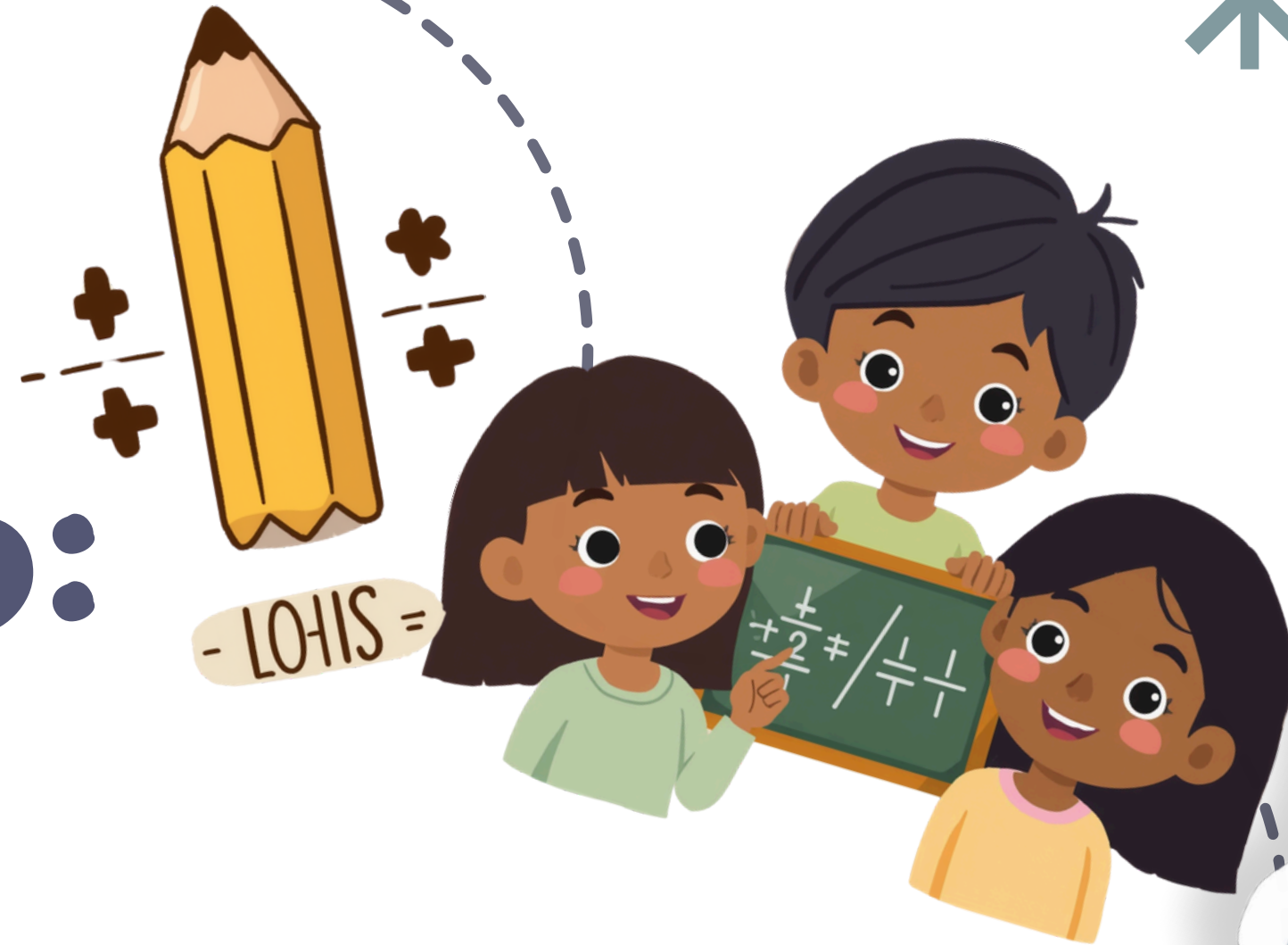
$$4 \times 3 = 4 + 0$$

1. Divide: $7 \div 4 = 1$, remainder 3
2. Bring down 8 $\rightarrow 38 \div 4 = 9$, remainder 2
3. Bring down 9 $\rightarrow 29 \div 4 = 7$, remainder 1
4. Answer: 197 R1

Step-by-Step:

$523 \div 6$

1. Divide: $5 \div 6 = 0$, bring down 2 $\rightarrow 52$
2. $52 \div 6 = 8$, write 8. Multiply: $8 \times 6 = 48$
3. Subtract: $52 - 48 = 4$, bring down 3 $\rightarrow 43$
4. $43 \div 6 = 7$, write 7. Multiply: $7 \times 6 = 42$
5. Subtract: $43 - 42 = 1$
6. Answer: 87 remainder 1



Practice Problems

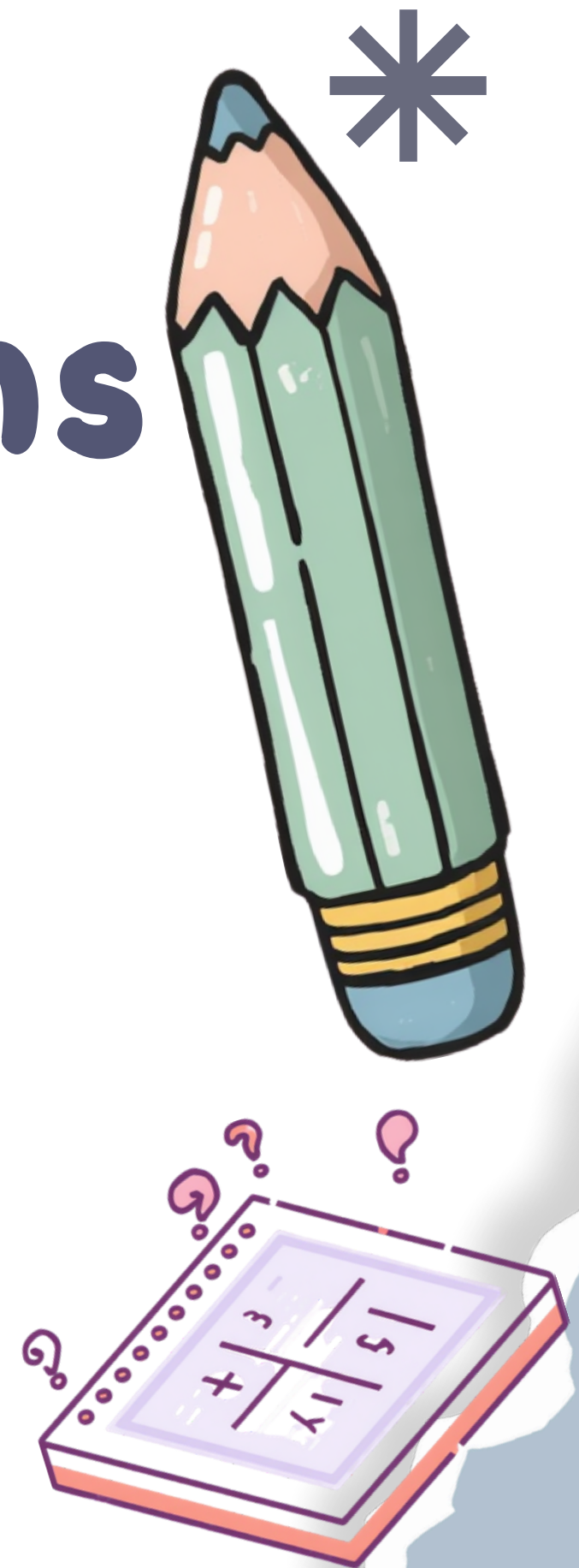


Try these division problems on your own!

• $456 \div 7 = ?$

• $789 \div 4 = ?$

• $523 \div 6 = ?$

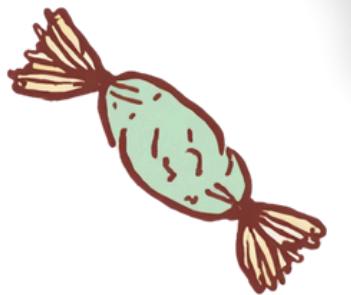




Real-World Uses of Division with Remainders

Division with remainders helps us in everyday life!

- Sharing sweets among friends fairly
- Packing items into boxes evenly
- Splitting money with leftovers to save





Tips and Tricks for Long Division



Break the problem into small steps.
Check your multiplication carefully.
Remember: the remainder is what's
left after dividing. Use estimation to
guess answers first!



Keep Practicing and You'll Be a Division Star!

Try these fun challenges and impress your
friends! Thank you for learning with us
today!

